MODELLING HUMAN ACTIVITY STATES

Using Hidden Markov Models and Smartphone Motion Sensors

1. Background and motivation

Smartphones have accelerometers and gyroscopes that record movement, but the sensor readings do not directly tell us what a person is doing. In this project, I used these sensors to recognize four activities: standing, walking, jumping, and staying still. This can support simple context-aware applications, such as a phone understanding whether its user is moving or resting. I used a Hidden Markov Model (HMM) because sensor readings form a sequence and an activity usually continues across more than one time window.

2. Data collection and dataset overview

I collected the data using SensorLogger on one iPhone 13 Pro. The project contains 52 labelled recordings: 13 for each activity. Each recording is about 5-10 seconds long. The four activities together contain 18,594 synchronized sensor rows and 371.79 seconds of recorded data. This is above the facilitator's clarification of 50 files in total and the rubric's 90-second minimum for each activity.

Activity	Files	Duration (s)	Samples	Median Hz
Standing	13	96.59	4,830	49.870
Walking	13	91.63	4,583	49.875
Jumping	13	91.21	4,562	49.875
Still	13	92.36	4,619	49.871
Total	52	371.79	18,594	49.873

Table 1. Number of recordings, duration, sensor samples, and measured sampling rate for each activity.

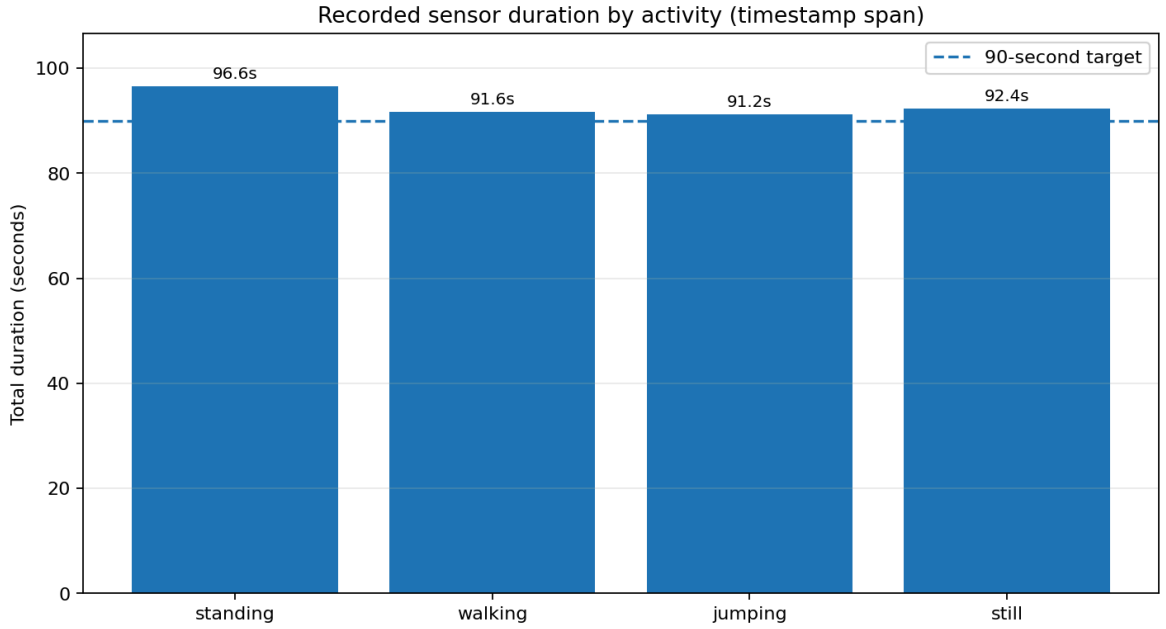


Figure 1. Total sensor recording time by activity. The dashed line shows the 90-second minimum.

3. Preprocessing and windowing

I used the calibrated Accelerometer.csv and Gyroscope.csv files for the model. The two streams were joined using matching timestamps. The quality checks found no missing or non-finite values in the selected sensor columns, and the accelerometer and gyroscope timestamps matched in all 52 recordings. The requested sample interval was 20 ms (about 50 Hz). The measured median rate was 49.87 Hz overall, with the activity medians all close to 49.87 Hz. Since the rates were already very close and timestamps matched, I kept the original samples and did not need to resample them. The combined dataset keeps the recording ID, activity, timestamp, elapsed time, and six sensor axes.

Window size

I used 2-second windows with a 1-second hop (50% overlap). At about 50 Hz, each window contains around 100 sensor samples. Two seconds gives enough points to calculate statistics and FFT features, while the 1-second hop produces several ordered observations from each short recording. The overlap increases the number of windows, but neighboring windows are related.

Example sensor signals

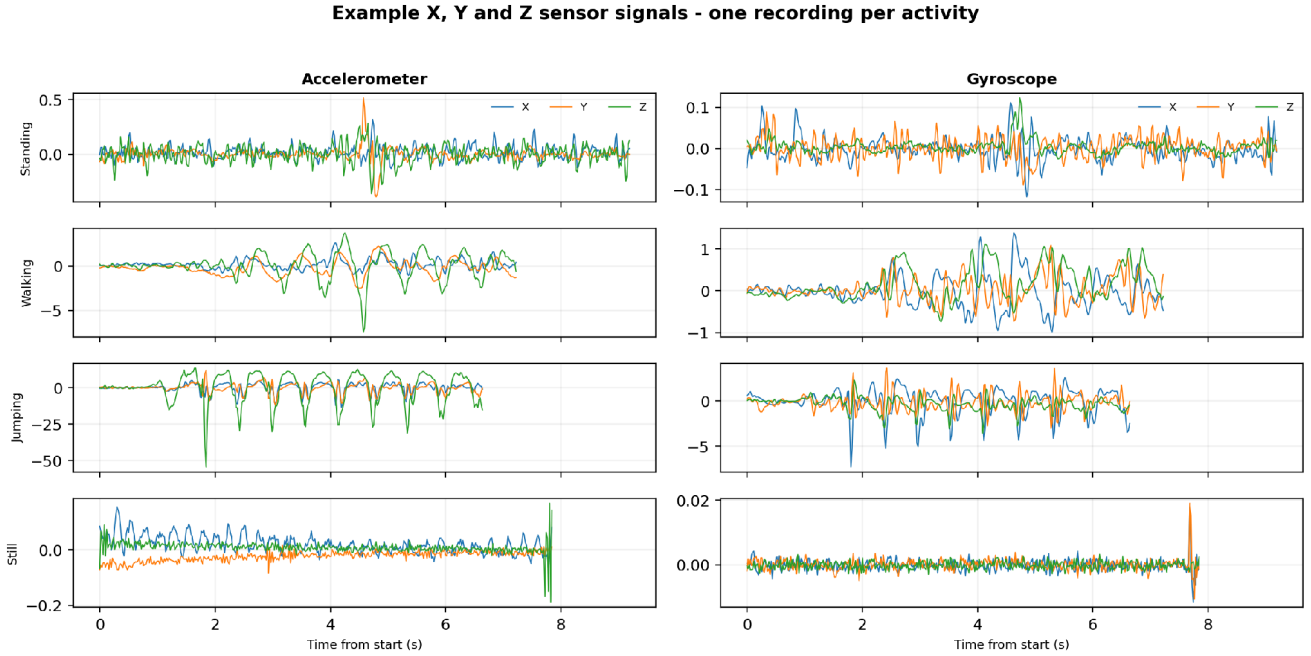


Figure 2. Actual X, Y, and Z accelerometer and gyroscope readings from one representative recording for each activity, generated from the notebook dataset.

The plots show different signal patterns: walking and jumping have larger and more repeated changes, while standing and still are generally less active. These are examples from individual recordings, not averages across all files.

4. Feature extraction

The feature notebook calculated 42 numeric features for every 2-second window. These include mean, standard deviation, variance, and RMS for the six sensor axes; accelerometer and gyroscope magnitude statistics; signal magnitude area (SMA); correlations between axes; and FFT-based frequency features. I selected 12 features for the HMM.

Feature group	Examples in the feature set	Why it helps
Time-domain	Mean, standard deviation, variance, RMS	Describe average level, variation, and movement strength.
Magnitude and SMA	Acceleration and gyroscope magnitude; SMA	Summarize movement across the three sensor axes.
Axis correlation	X-Y, X-Z, and Y-Z correlations	Describe how movement is shared between axes.
Frequency-domain (FFT)	Dominant frequency and spectral energy	Show repeated motion patterns and signal energy.

Table 2. Time-domain and frequency-domain features are extracted, and their purpose is explained.

The 12 HMM inputs were acceleration magnitude mean, standard deviation, RMS, dominant frequency, and spectral energy; the same five features for gyroscope magnitude; and accelerometer and gyroscope SMA. Mean describes the average magnitude, standard deviation and RMS describe variation and intensity, dominant frequency can show repeated movement such as steps or jumps, spectral energy describes the amount of signal energy, and SMA summarizes total three-axis activity.

I used Z-score normalization because the features have different numeric scales. The means and standard deviations were fitted using training windows only, then applied to the test windows. This prevents information from the test set from being used when fitting the scaler. The 52 recordings produced 292 windows: 247 training windows and 45 test windows.

5. HMM setup and implementation

HMM element	Meaning	Use in this project
Hidden states (Z)	The activity at each time step.	Standing, walking, jumping, and still.
Observations (X)	Feature vector for each window.	12 normalized features in time order.
Initial probabilities (π)	Chance of starting in each state.	Estimated from labelled training sequences with small smoothing.
Transition matrix (A)	Chance of moving between states.	Estimated from training sequences and refined by Baum-Welch.
Emissions (B)	Chance of features given a state.	Diagonal-covariance Gaussian distributions.

Table 3. HMM components and how each was set up in this project.

Algorithms used

I initialized the Gaussian emission means and variances using the training labels, then used Baum-Welch (an expectation-maximization algorithm) to refine the transition and emission parameters. The model used diagonal covariance, so it estimates a separate variance for each feature instead of every feature-to-feature covariance. Training stopped when the change in total log-likelihood was below 1e-5; it converged after 7 iterations. I used Viterbi to find the most likely activity-state sequence. Sensitivity, specificity, accuracy, and the confusion matrix were used to evaluate its predictions.

6. Results and evaluation on unseen recordings

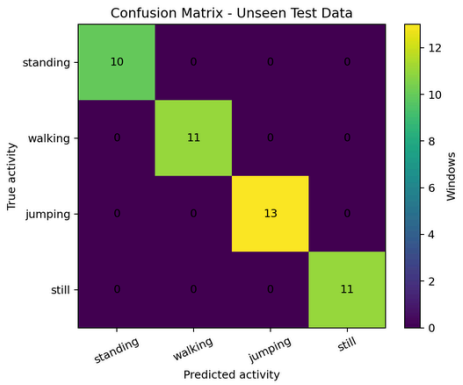
I set aside two complete recordings per activity before training. The other 44 recordings were used for training. These eight test recordings were not used to fit the model or the scaler. They were separate files from the collected dataset, made by the same participant using the same phone; this tests new recordings from this dataset, not a different participant or device. The test set contained 45 windows.

Activity	Test windows	Sensitivity	Specificity	Accuracy
Standing	10	100.0%	100.0%	100.0%
Walking	11	100.0%	100.0%	100.0%
Jumping	13	100.0%	100.0%	100.0%
Still	11	100.0%	100.0%	100.0%

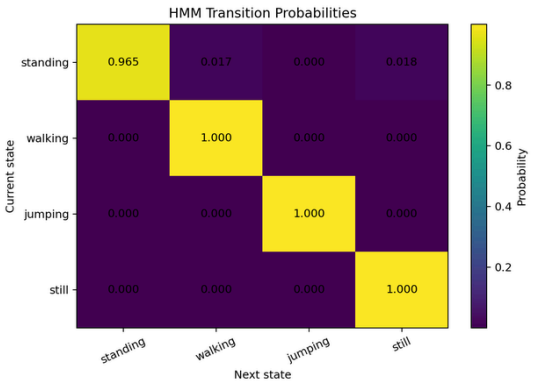
Table 4. Per-activity results for the 45 Viterbi-decoded test windows.

The model correctly classified all 45 test windows, giving 100% accuracy on this split. There were no off-diagonal errors in the confusion matrix. The emission likelihood plot also shows high likelihood for the matching activity in these test windows. The decoded sequences match the true labels for the eight held-out recordings.

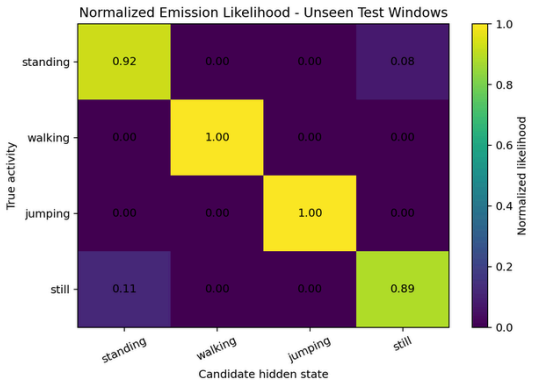
Confusion matrix



Transition probabilities



Emission likelihood



Viterbi decoded sequences

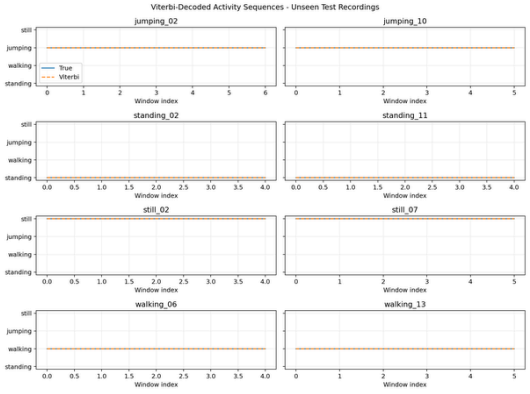


Figure 3. HMM results: confusion matrix, transition probabilities, normalized emission likelihoods and Viterbi-decoded test sequences.

The transition matrix is strongly diagonal because each training recording contains only one activity. It shows that a state tends to continue across adjacent windows. It does not yet show natural changes such as standing followed by walking.

7. Discussion and conclusion

What the results show

In this test, the HMM separated all four activities correctly. Walking and jumping generally showed greater and more repeated changes in the sensor signals. Standing and still had smaller changes and may be harder to separate in a larger dataset. The result is encouraging, but 100% accuracy on 45 windows does not mean the model will always be correct in real situations.

The high diagonal values in the transition matrix make sense because each recording contains one activity, and the activity continues across its windows. However, the dataset does not contain continuous recordings that move naturally between different activities. Mixed-activity recordings would be needed to learn those transitions.

Limitations and improvements

- The data came from one participant and one iPhone 13 Pro. The test recordings were held out from this dataset, not collected from a new participant or phone.
- The test set was small: 8 recordings and 45 windows. The result should not be treated as guaranteed performance for other people or environments.
- The windows overlap by 50%, so neighboring windows are related, even though complete recordings were kept separate between training and testing.
- Phone placement and orientation can affect sensor readings. Although there were no missing values and timestamps were aligned, small sensor noise may remain. I did not apply a separate filtering step.
- Future work could collect data from more people and phones, use longer mixed-activity sessions, test on a separate session, and compare filtering or additional features.

Conclusion

In this project, I collected smartphone accelerometer and gyroscope data, checked and combined the recordings, extracted time- and frequency-domain features, normalized them, and built a Gaussian HMM. I used Baum-Welch to train the model and Viterbi to predict activity sequences. The model achieved 100% accuracy on the 45 held-out windows, but more varied data is needed to test how well it works beyond this collection.

Video evidence

Activity	Video evidence
Standing	https://drive.google.com/file/d/1727hAD6apr7gdVpzpGksJUpqn2bk_koq/view?usp=sharing
Walking	https://drive.google.com/file/d/101l9eTLRyI724kvgGCErPI5OIb1_1_3p/view?usp=sharing
Jumping	https://drive.google.com/file/d/1FqYdWWoW70qHXB5u_EZz8S1zGi3FFL6B/view?usp=sharing
Still (no movement)	https://drive.google.com/file/d/1727hAD6apr7gdVpzpGksJUpqn2bk_koq/view?usp=sharing